AQ3.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

Consider the following statements:

- Statement 1: $0(0,0)+0(1,1)=(0,0)$ implies that the set $\{(0,0),(1,1)\}$ is a linearly independent subset of R^2.
- Statement 2: $2(1,0)+2(0,1)=(2,2)$ implies that the set $\{(1,0),(0,1),(2,2)\}$ is a linearly dependent subset of R^2.
- Statement 3: $2(1,0)+2(0,1)=(2,2)$ implies that the set $\{(1,0),(2,2)\}$ is a linearly dependent subset of R^2.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
If $\alpha(1,1,0)+\beta(0,1,1)+\gamma(1,0,1)=v$ for some $\alpha,\beta,\gamma\in R$ and $v\in R^3$, then choose the set of correct options.

- [ ] If $v=(0,0,0)$, then one of the possible solutions of $\alpha,\beta,$ and γ is $\alpha=\frac{1}{2}=\gamma$, and $\beta=-\frac{1}{2}$.
- [ ] If $v=(0,0,0)$, then one of the possible solutions of $\alpha,\beta,$ and γ is $\alpha=\beta=\gamma=0$.
- [ ] If $v=(0,0,0)$, then the possible solution of $\alpha,\beta,$ and γ is unique.
- [ ] If $v=(1,0,0)$, then one of the possible solutions of $\alpha,\beta,$ and γ is $\alpha=\frac{1}{2}=\gamma$, and $\beta=-\frac{1}{2}$.
- [ ] The set $\{(1,1,0),(0,1,1),(1,0,1)\}$ is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $v=(0,0,0)$, then one of the possible solutions of $\alpha,\beta,$ and γ is $\alpha=\beta=\gamma=0$.
If $v=(0,0,0)$, then the possible solution of $\alpha,\beta,$ and γ is unique.
If $v=(1,0,0)$, then one of the possible solutions of $\alpha,\beta,$ and γ is $\alpha=\frac{1}{2}=\gamma$, and $\beta=-\frac{1}{2}$.
The set $\{(1,1,0),(0,1,1),(1,0,1)\}$ is linearly independent.
1 point

Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$ in R^2. Choose the set of correct options.

☐ $\{(1,2),(-1,2)\}$ is a linearly dependent set.

☐ $\{(-1,2),(-3,1)\}$ is a linearly independent set.

☐ $\{(-3,1),(-1,2),(3,1)\}$ is a linearly dependent set.

☐ $\{(3,1),(-3,1)\}$ is a linearly dependent set.

☐ $\{(3,1),(1,2),(-3,1)\}$ is a linearly dependent set.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{(-1,2),(-3,1)\}$ is a linearly independent set.
$\{(-3,1),(-1,2),(3,1)\}$ is a linearly dependent set.
$\{(3,1),(1,2),(-3,1)\}$ is a linearly dependent set.

1 point

Let V be a vector space and non zero vectors $v_1, v_2, v_3,$ and $v_4 \in V$. If v_1 is a linear combination of v_i, for $i = 2, 3, 4$, then which of the following is (are) correct?

☐ The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.

☐ The set $\{v_2, v_3, v_4\}$ must be linearly dependent.

☐ The set $\{v_2, v_3, v_4\}$ must be linearly independent.

☐ The set $\{v_1\}$ must be linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.
The set $\{v_1\}$ must be linearly independent.

1 point

Choose the correct set of options.

☐ Union of two distinct linearly independent sets may be linearly independent.
☐ Union of two distinct linearly independent sets must be linearly dependent.
☐ Non-empty intersection of two linearly independent sets must be linearly independent.
☐ Non-empty intersection of two linearly independent sets must be linearly dependent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Union of two distinct linearly independent sets may be linearly independent.
Non-empty intersection of two linearly independent sets must be linearly independent.

If vectors v_1, v_2, v_3 are linearly independent and $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$, then find the value of $\alpha + \beta + \gamma$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point

Consider the set of matrices $S = \left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\x\end{bmatrix}\right\} \subset M_{3\times 1}(\mathbb{R})$. If $x \in \{9,4,2,0\}$, then find the maximum value of x so that S is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4

1 point

Level 2

1 point

Let S be a linearly independent subset of $\mathbb{R}^3$. Choose the correct set of options.

- [ ] $S \cup \{(1,0,0)\}$ must be linearly independent.
- [ ] $S \cup \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
- [ ] $S \setminus \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
- [ ] There may exist some $v \in \mathbb{R}^3$, such that $S \cup \{v\}$ is still linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$S \setminus \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
There may exist some $v \in \mathbb{R}^3$, such that $S \cup \{v\}$ is still linearly independent.

1 point

Consider a set of vectors $S = \{(1,1,-1), (-1,1,1), (0,\frac{1}{2},0), (0,1,-2), (1,0,-2)\}$. Choose the set of correct options.

- [ ] The singleton set $\{(0,1,-2)\}$ is linearly dependent.
- [ ] If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.
- [ ] The set $\{(-1,1,1), (0,\frac{1}{2},0), (0,1,-2)\}$ is a linearly independent set of vectors.
- [ ] The set S is a linearly independent set of vectors.
- [ ] The set $\{\alpha, \beta, \gamma\}$ is a linearly dependent set of vectors for any $\alpha, \beta, \gamma \in S$, where

all the three are distinct vectors.

☐

The set $\{\alpha, \beta, \gamma, \delta\}$ is a linearly independent set of vectors for any $\alpha, \beta, \gamma, \delta \in S$, where all the four are distinct vectors.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

1 point

Let S be the solution set of a system of homogeneous linear equations with 3 variables and 3 equations, whose matrix representation is as follows:

$$Ax = 0$$

where A is the 3×3 coefficient matrix and x denotes the column vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Choose the set of correct options.

☐

If v_1 and v_2 are in S, then any linear combination of v_1 and v_2 will also be in S.

☐

The set S will be a subspace of R^3, with respect to usual addition and scalar multiplication as in R^3.

☐

The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S.

☐

The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If v_1 and v_2 are in S, then any linear combination of v_1 and v_2 will also be in S.

The set S will be a subspace of R^3, with respect to usual addition and scalar multiplication as in R^3.

The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S.

The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2.

Check Answers

Your score is: 0/10